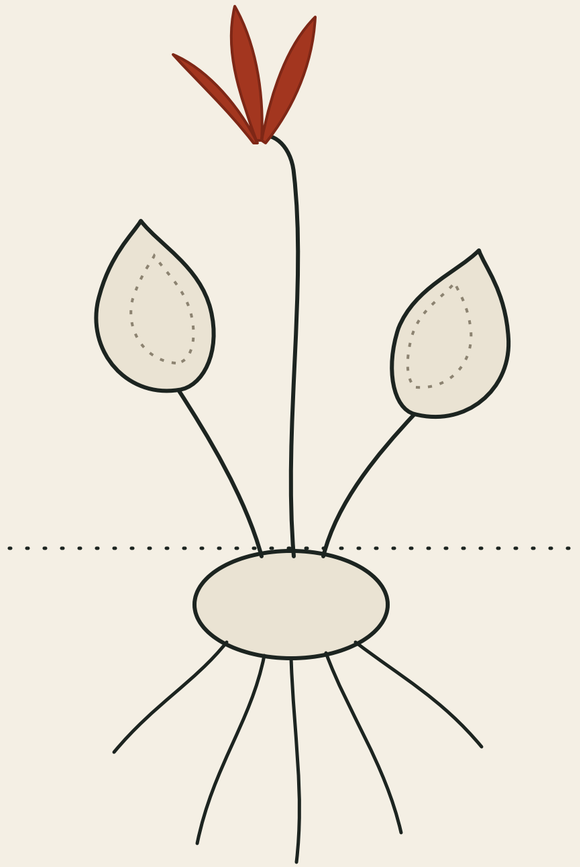
ETHNOHACK 2026 · TRACK 1, TRADITIONAL KNOWLEDGE MINING

Rhizotomos

ῥιζοτόμος, "root-cutter": the ancient Greek word for a herbalist

Which plant did Dioscorides mean?

Claude repeats, step by step, a published expert method for identifying the plants of *De materia medica*, and every step is scored against the experts' own data.



Sanchay Singh · Ashank Kumar

github.com/ashank-crazy/rhizotomos

One name, five species

κυκλάμινος

Dioscorides, *De materia medica* 2.164

“Its leaves are like ivy, mottled above and below with whitish spots; a bare stem four fingers long; rose-like, purplish flowers; a black root, somewhat flat.”

Species proposed before	Times proposed
Cyclamen persicum	3
Cyclamen graecum	2
Cyclamen cyprium	2
Cyclamen hederifolium	1
Cyclamen purpurascens	also proposed

Every study that starts from an ancient remedy inherits this choice. The usual answers come from dictionaries, and many of them are doubtful.

The experts’ method is a line-up

Witness statement

Dioscorides’ description, coded into 36 characters from height to habitat.

Suspects

Species proposed before, each described from Flora of Turkey or of Cyprus.

Evidence

Each statement against the flora: **good**, **partial**, **no** or **absent**.

Verdict

Organs agreeing minus disagreeing:**STRONG** ≥ 5, **MODERATE** 3–4, **WEAK** ≤ 2.

Lardos et al. (2024) did this by hand for 61 plants and published all 1,186 calls and every verdict under CC BY. That is an answer key.

Claude takes each step, and each step is checked

A Read

Claude sorts each Greek chapter into the 36 characters, quoting the text.

699 / 699
quotes found by code in their chapter

B Match, blind

Each statement against each flora entry, species names hidden.

1,186 calls
by the experts, compared one by one

C Score

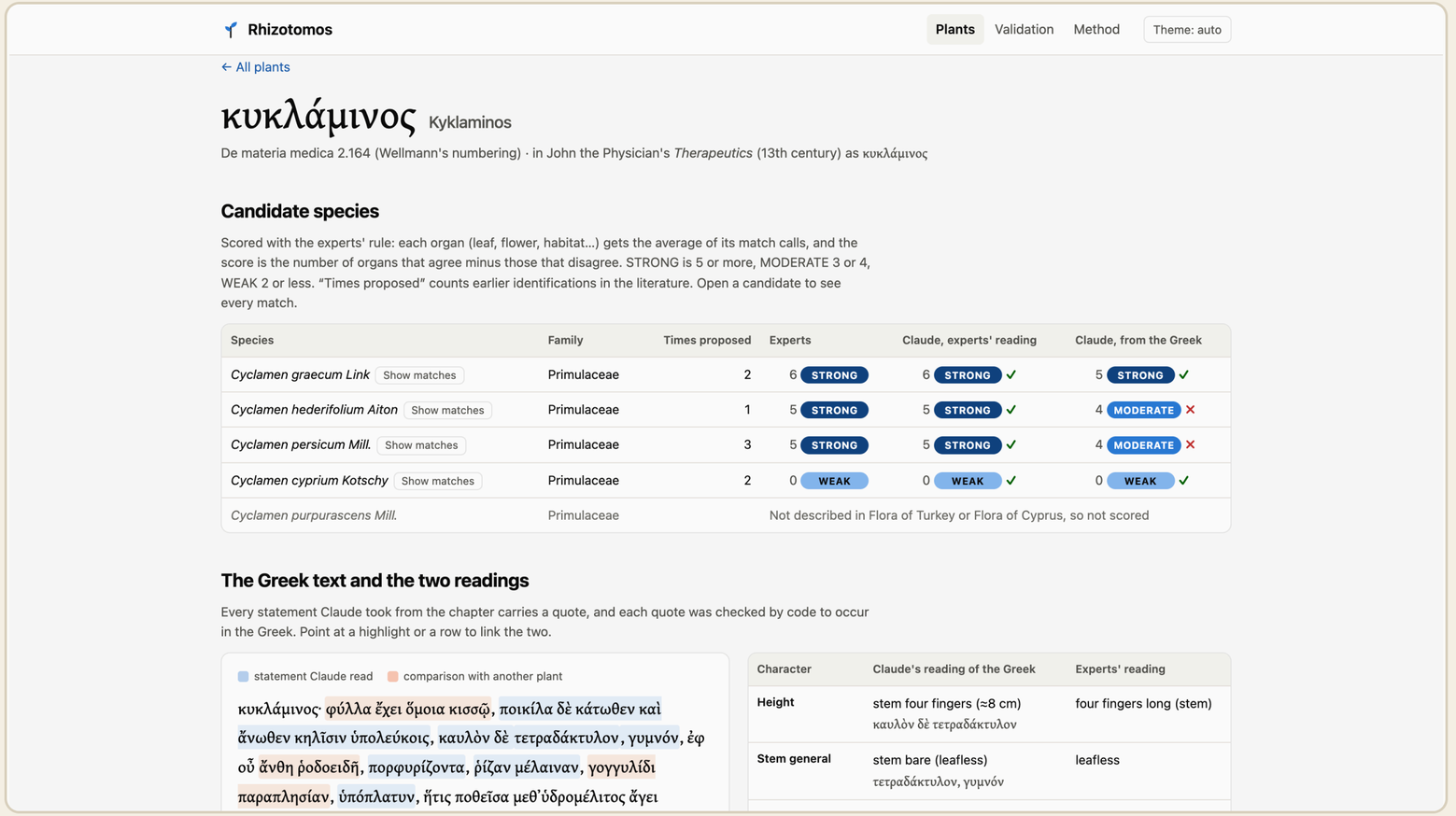
The experts’ own Stage 5 rule, in code, turns calls into verdicts.

117 / 120
published verdicts reproduced

Pass E experts’ reading, Claude’s matching

Pass A Claude’s reading and matching, end to end

Every plant, every reading, every call



EXPERTS

C. graecum first, scoring 6, then 5, 5 and 0.

PASS E

Claude's calls give the same four scores.

PASS A

From the Greek: 5, 4, 4, 0. *C. graecum* still first.

ashank-crazy.github.io/rhizotomos

Claude matches three expert verdicts in four

Pass E

the experts' reading, Claude's matching

76%

same verdict, 122 candidates

68–84%

88%

same best candidate, 31 plants

77–96%

Pass A

Claude from the Greek, end to end

60%

same verdict, 122 candidates

51–69%

80%

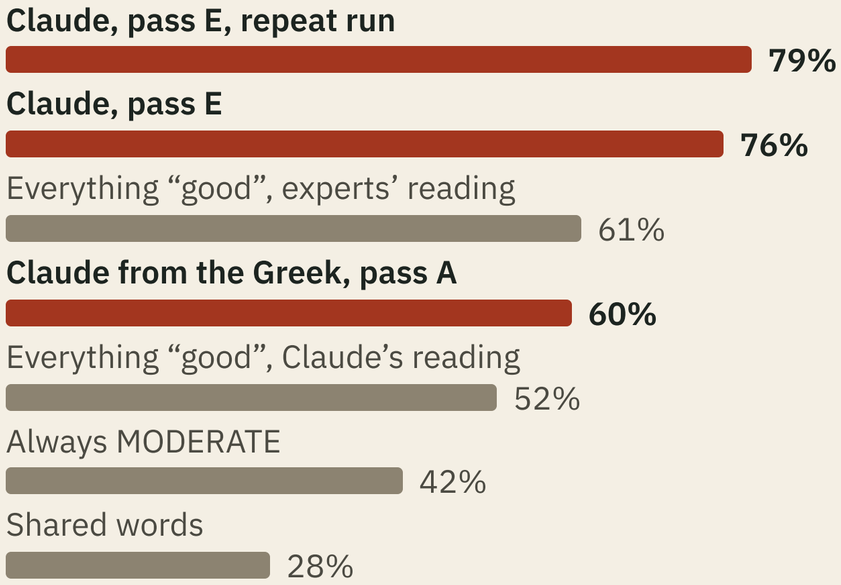
same best candidate, 31 plants

67–91%

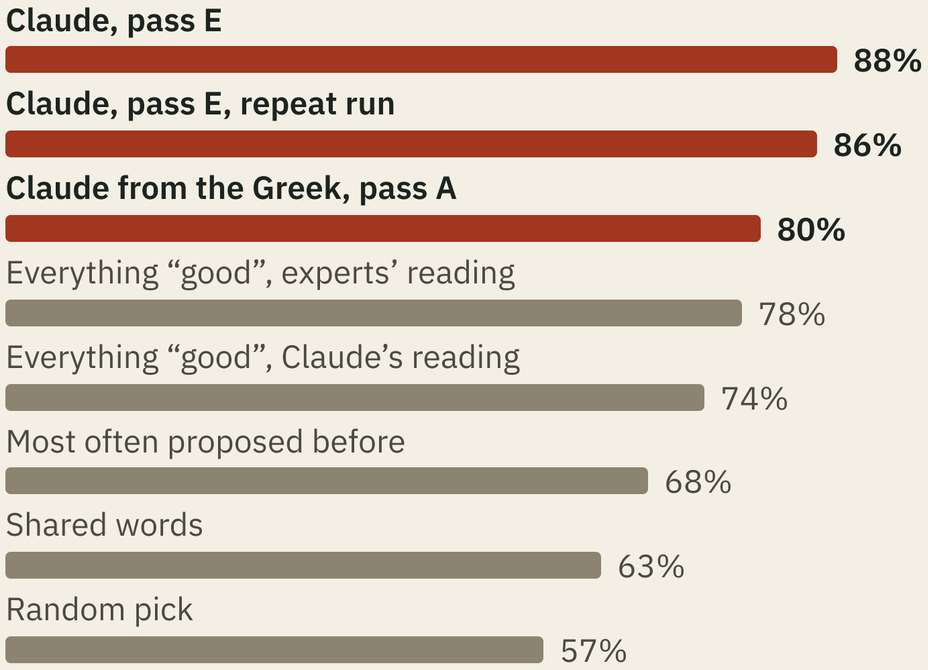
Ranges are 95% bootstrap intervals over plants. On the 54 plants never looked at during development: 78% and 91% for pass E, 61% and 84% for pass A.

Calling everything “good” is the bar to beat

Same verdict, 122 candidates



Same best candidate, 31 plants



Claude in red. Its matching adds 15 points over everything “good”, and 8 on its own reading.

Six checks behind the numbers

699 / 699

Greek quotes found by code in their chapter

117 / 120

published verdicts the scoring code reproduces

Blind

anonymous labels, shuffled, no species names in view

78% · 91%

pass E on the 54 plants never seen in development

88% · 98%

same calls and best candidates when pass E is redone

1 command

rebuilds every number offline, checked by CI

What this does not show

Reading is the harder step	Claude’s reading covers 83% of the organs the experts coded, which is where pass A loses ground.
The modern side is the experts’	Flora descriptions cover only characters in their reading. Claude extracting its own is untested.
A small benchmark	61 plants and 122 scored candidates, so the intervals are wide.
Botanical stage only	The experts’ final identifications also weigh medicinal uses and expert review.
No medical claims	Identifications are hypotheses about historical texts, not advice on any plant.

From 61 plants to 827 chapters

827 chapters

Wellmann’s whole text is already ingested, with its synonym lists.

Triage

Score every chapter, and send experts the close calls first.

Other herbals

The same steps fit John the Physician, Galen and later texts.

Thank you

Sanchay Singh · Ashank Kumar

github.com/ashank-crazy/rhizotomos · ashank-crazy.github.io/rhizotomos

Built with Claude (Opus 5.5) in Claude Code. Data: Lardos et al. 2024, CC BY 4.0; Wellmann’s Dioscorides via First1KGreek, CC BY-SA 4.0.